TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	63.0 ± 3.61
Correlation Stability	88.3 ± 0.90
Distribution Consistency	75.0 ± 3.55
Entropy Preservation	67.4 ± 1.43
Mean Z Score Drift	0.5 ± 0.03
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	5.7 ± 2.41
Data Density	100.0 ± 0.00
Coverage Score	88.2 ± 2.93
Feature Accuracy	53.7 ± 2.00
Novelty Score	0.7 ± 0.03

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Dataset Audit: heart_failure_clinical_records.csv

1. Dataset Portrait

Attribute	Value
Rows	299
Columns	13
Numeric features	13
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
age	60.834	65.236	+4.402	7.2%	OK
anaemia	0.431	0.232	-0.199	46.2%	Drift
creatinine_phosphokinase	581.839	713.616	+131.777	22.6%	Warn
diabetes	0.418	0.152	-0.267	63.8%	Drift
ejection_fraction	38.084	33.152	-4.932	13.0%	Warn
high_blood_pressure	0.351	0.202	-0.149	42.5%	Drift
platelets	263358.029	248392.630	-14965.400	5.7%	OK
serum_creatinine	1.394	1.866	+0.473	33.9%	Drift
serum_sodium	136.625	134.646	-1.979	1.4%	OK
sex	0.649	0.384	-0.265	40.8%	Drift
smoking	0.321	0.020	-0.301	93.7%	Drift
time	130.261	67.394	-62.867	48.3%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	63.0 ± 3.61
Correlation Stability (*2)	88.3 ± 0.90
Distribution Consistency (*3)	75.0 ± 3.55
Entropy Preservation (*4)	67.4 ± 1.43
Mean Z Score Drift (*5)	0.5 ± 0.03
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	5.7 ± 2.41

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	88.2 ± 2.93
Feature Accuracy (*10)	53.7 ± 2.00
Novelty Score (*11)	0.7 ± 0.03

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.805	N/A	0.78	-0.025	
F1	0.6894302795561282	N/A	0.7104858950347672	0.021055615478639045	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7499999999999999	N/A	0.7849999999999999	0.03499999999999996	
F1	0.5667171896628354	N/A	0.7052746879637636	0.13855749830092828	*

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6133333333333335	N/A	0.5349999999999999	-0.07833333333333332	*
F1	0.007142857142857143	N/A	0.4356021459961791	0.42845928885332196	*

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6333333333333334	N/A	0.6066666666666667	-0.026666666666666665	
F1	N/A	N/A	0.27266383570606173	0.27266383570606173	*

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.4966666666666667	N/A	0.4716666666666667	-0.024999999999999994	
F1	0.2705077634609716	N/A	0.38420975203703	0.1137019885760584	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.

ACCURACY	0.825	N/A	0.8266666666666668	0.00166666666666607	
F1	0.7417404771442427	N/A	0.7535785909067643	0.011838113762521552	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.835	N/A	0.8433333333333334	0.00833333333333337	
F1	0.7547418227770806	N/A	0.7773865902636359	0.02264476748655523	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8283333333333335	N/A	0.8316666666666667	0.00333333333333332 14	
F1	0.7520295309201247	N/A	0.7668276318535107	0.014798100933385983	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7833333333333334	N/A	0.7783333333333333	-0.005000000000000027	
F1	0.6709910607031591	N/A	0.6883587562198367	0.017367695516677707	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8333333333333334	N/A	0.8383333333333333	0.005000000000000044	
F1	0.7508527868346062	N/A	0.7729466210481248	0.022093834213518597	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	75.0	0.0	582.0	0.0	20.0	1.0	265000.0	1.9	130.0	1.0
2	55.0	0.0	7861.0	0.0	38.0	0.0	263358.03	1.1	136.0	1.0
3	65.0	0.0	146.0	0.0	20.0	0.0	162000.0	1.3	129.0	1.0
4	50.0	1.0	111.0	0.0	20.0	0.0	210000.0	1.9	137.0	1.0
5	65.0	1.0	160.0	1.0	20.0	0.0	327000.0	2.7	116.0	0.0
6	90.0	1.0	47.0	0.0	40.0	1.0	204000.0	2.1	132.0	1.0
7	75.0	1.0	246.0	0.0	15.0	0.0	127000.0	1.2	137.0	1.0
8	60.0	1.0	315.0	1.0	60.0	0.0	454000.0	1.1	131.0	1.0
9	65.0	0.0	157.0	0.0	65.0	0.0	263358.03	1.5	138.0	0.0
10	80.0	1.0	123.0	0.0	35.0	1.0	388000.0	9.4	133.0	1.0
11	75.0	1.0	81.0	0.0	38.0	1.0	368000.0	4.0	131.0	1.0
12	62.0	0.0	231.0	0.0	25.0	1.0	253000.0	0.9	140.0	1.0
13	45.0	1.0	981.0	0.0	30.0	0.0	136000.0	1.1	137.0	1.0
14	50.0	1.0	168.0	0.0	38.0	1.0	276000.0	1.1	137.0	1.0
15	49.0	1.0	80.0	0.0	30.0	1.0	427000.0	1.0	138.0	0.0
16	82.0	1.0	379.0	0.0	50.0	0.0	47000.0	1.3	136.0	1.0
17	87.0	1.0	149.0	0.0	38.0	0.0	262000.0	0.9	140.0	1.0
18	45.0	0.0	582.0	0.0	14.0	0.0	166000.0	0.8	127.0	1.0
19	70.0	1.0	125.0	0.0	25.0	1.0	237000.0	1.0	140.0	0.0
20	48.0	1.0	582.0	1.0	55.0	0.0	87000.0	1.9	121.0	0.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	46.27531112774819	0.0	1882.0	1.0	31.0	0.0	330174.06661675544	1.0744937774450363	136.0	0.0
2	49.56677767451597	0.0	482.0	0.0	28.0	1.0	319000.0	1.0433222325484033	131.0	0.0
3	75.10692863750042	1.0	578.0	0.0	30.0	0.0	225007.12857583337	1.8240832820583106	134.0	1.0
4	69.44014570834219	1.0	316.0	0.0	36.0	0.0	222160.0161898158	1.4640064759263196	141.0	1.0
5	73.52714814565266	0.0	914.0	1.0	33.0	0.0	263358.03	2.3180049144070862	135.0	0.0
6	60.69977141151256	0.0	1773.0	0.0	27.0	0.0	364020.3200238824	2.0440182870789934	143.0	0.0
7	45.0	1.0	66.0	1.0	25.0	0.0	233000.0	0.8	135.0	1.0
8	70.0	0.0	161.0	0.0	25.0	0.0	244000.0	1.2	142.0	0.0
9	58.0	1.0	400.0	0.0	40.0	0.0	164000.0	1.0	139.0	0.0
10	55.0	0.0	572.0	1.0	35.0	0.0	231000.0	0.8	143.0	0.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	77.33215078877097	0.0	582.0	0.0	31.0	0.0	263358.03	1.83	134.0	0.0
2	79.64357235983245	0.0	153.0	0.0	38.0	0.0	149047.52368535567	1.8845548022594059	143.0	0.0
3	57.6438696554128	0.0	512.0	0.0	30.0	0.0	99003.629654659116	1.4974821379290855	135.0	1.0
4	49.87775728033292	0.0	607.0	0.0	35.0	0.0	311100.1844770037	1.8022058242663381	135.0	1.0
5	45.94214823156065	0.0	582.0	0.0	20.0	0.0	163550.41459794233	0.7623140707375738	128.0	0.0
6	64.01336370263422	0.0	380.0	0.0	25.0	0.0	265723.5332847349	1.1934224246842282	136.0	1.0
7	60.0	0.0	2656.0	1.0	30.0	0.0	305000.0	2.3	137.0	1.0
8	45.0	1.0	130.0	0.0	35.0	0.0	174000.0	0.8	139.0	1.0
9	45.0	0.0	582.0	1.0	55.0	0.0	543000.0	1.0	132.0	0.0
10	65.0	1.0	59.0	1.0	60.0	0.0	172000.0	0.9	137.0	0.0

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	70.7772022542234	0.0	599.0	0.0	21.0	0.0	263358.03	1.77326292354416 9	134.0	1.0
2	55.9969922410112 06	0.0	822.0	1.0	25.0	0.0	71668.3376438826 7	1.89949870683520 14	136.0	0.0
3	63.4089784039375 4	0.0	578.0	0.0	31.0	0.0	162684.262180621 56	0.89797729786527 87	136.0	0.0
4	53.4921024829728 9	0.0	1708.0	0.0	23.0	0.0	315256.581264189 27	1.1	137.0	1.0
5	58.1117440447324 26	0.0	61.0	0.0	37.0	0.0	153143.670914655 96	5.72816454267201 3	133.0	1.0
6	54.2183167379814 75	1.0	112.0	0.0	18.0	0.0	206890.841631009 27	0.89272277540061 75	138.0	0.0
7	50.0	0.0	582.0	1.0	38.0	0.0	310000.0	1.9	135.0	1.0
8	79.0	1.0	55.0	0.0	50.0	1.0	172000.0	1.8	133.0	1.0
9	65.0	1.0	160.0	1.0	20.0	0.0	327000.0	2.7	116.0	0.0
10	70.0	0.0	92.0	0.0	60.0	1.0	317000.0	0.8	140.0	0.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	58.4258421249398 05	0.0	528.0	0.0	50.0	0.0	125089.397961136 9	0.78431529648305 82	134.0	0.0
2	70.0435044828257 8	0.0	127.0	0.0	25.0	1.0	236956.495517174 22	1.03262836211933 17	139.0	0.0
3	68.8767012003207 8	0.0	463.0	1.0	46.0	1.0	185673.979279807 54	1.15506804801283 12	139.0	1.0
4	45.0	0.0	7198.0	0.0	25.0	0.0	389646.310300509 9	1.0	139.0	1.0
5	69.2183037777744 8	0.0	547.0	0.0	20.0	0.0	266582.143407398 6	1.18544641481503 43	134.0	1.0
6	79.4472335019760 9	0.0	792.0	0.0	37.0	0.0	263358.03	1.14035195435574 59	134.0	1.0
7	48.0	1.0	131.0	1.0	30.0	1.0	244000.0	1.6	130.0	0.0
8	95.0	1.0	371.0	0.0	30.0	0.0	461000.0	2.0	132.0	1.0
9	45.0	1.0	981.0	0.0	30.0	0.0	136000.0	1.1	137.0	1.0
10	50.0	0.0	582.0	1.0	38.0	0.0	310000.0	1.9	135.0	1.0

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	64.6848063402939	1.0	246.0	1.0	25.0	0.0	197873.92253611758	1.4693426051353398	129.0	0.0
2	58.05801240051275	0.0	186.0	0.0	21.0	0.0	121200.91927941889	2.6669614880615304	128.0	1.0
3	82.53993661248599	1.0	88.0	0.0	32.0	0.0	198848.01448857464	0.8931994566784512	138.0	0.0
4	51.41712473478663	0.0	287.0	0.0	37.0	0.0	216531.421775545	2.1051453489668384	132.0	0.0
5	73.85467957999738	0.0	301.0	0.0	45.0	0.0	230548.769008006 3	2.2403448588003 67	138.0	0.0
6	84.33040713684436	0.0	28.0	0.0	45.0	0.0	359598.24428210664	2.9241128088423602	132.0	0.0
7	78.0	0.0	224.0	0.0	50.0	0.0	481000.0	1.4	138.0	1.0
8	70.0	0.0	582.0	0.0	20.0	1.0	263358.03	1.83	134.0	1.0
9	61.0	0.0	248.0	0.0	30.0	1.0	267000.0	0.7	136.0	1.0
10	40.0	0.0	478.0	1.0	30.0	0.0	303000.0	0.9	136.0	1.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	74.95877675233974	0.0	582.0	0.0	29.0	1.0	263358.03	1.83	134.0	0.0
2	56.42945707279832	1.0	204.0	0.0	50.0	0.0	152214.72853639917	7.05643434176134	135.0	0.0
3	60.71332722595645 5	0.0	515.0	1.0	36.0	0.0	457562.6104787994	0.7754784975852872	137.0	1.0
4	50.04555074701113 4	1.0	130.0	0.0	29.0	1.0	243349.14294413282	1.5442122523542419	130.0	0.0
5	47.12711218703093	1.0	954.0	0.0	29.0	0.0	136243.09853566068	1.3005562919200593	136.0	1.0
6	59.08357055825574	0.0	557.0	0.0	40.0	0.0	216908.35705582556	3.5716998781558 03	133.0	0.0
7	81.0	0.0	4540.0	0.0	35.0	0.0	231000.0	1.18	137.0	1.0
8	70.0	0.0	2695.0	1.0	40.0	0.0	241000.0	1.0	137.0	1.0
9	70.0	0.0	835.0	0.0	35.0	1.0	305000.0	0.8	133.0	0.0
10	80.0	0.0	776.0	1.0	38.0	1.0	192000.0	1.3	135.0	0.0

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	68.2032323759734	0.0	1755.0	0.0	27.0	0.0	250302.942445228	0.92697057554771	137.0	1.0
	4						92	07		
2	71.1927370133137	0.0	582.0	0.0	20.0	1.0	263749.717678750	1.84669831818639	133.0	1.0
	3						17	23		
3	80.0469946703826	0.0	72.0	1.0	32.0	0.0	198621.408002623	1.17702346671038	132.0	1.0
	5						04	4		
4	68.4466859479450	0.0	556.0	0.0	20.0	0.0	265861.671486986	1.2	134.0	1.0
	8						3			
5	63.6981769512490	0.0	222.0	1.0	52.0	0.0	189562.187658501	1.02603646097501	140.0	0.0
	9						1	83		
6	72.7761698813017	0.0	1219.0	0.0	26.0	0.0	243814.922007913	1.16298440158264	140.0	0.0
							2	4		
7	75.0	0.0	582.0	1.0	30.0	1.0	263358.03	1.83	134.0	0.0
8	53.0	0.0	207.0	1.0	40.0	0.0	223000.0	1.2	130.0	0.0
9	60.0	1.0	2281.0	1.0	40.0	0.0	283000.0	1.0	141.0	0.0
10	59.0	1.0	280.0	1.0	25.0	1.0	302000.0	1.0	141.0	0.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	62.1245909437629	0.0	232.0	0.0	24.0	1.0	253012.459094376	0.90498363775051	139.0	1.0
	25						3	7		
2	69.7706164557680	1.0	135.0	1.0	33.0	0.0	222737.196202543	1.0	135.0	1.0
	8						24			
3	82.2964666777076	0.0	1063.0	0.0	41.0	0.0	263358.03	2.53794463891511	135.0	0.0
	6							23		
4	82.4106609955912	1.0	268.0	0.0	47.0	0.0	224248.614075905	1.46703624642515	140.0	1.0
	9						67	14		
5	54.7573723974709	0.0	1947.0	0.0	25.0	0.0	313805.897917976	1.13882041640464	137.0	1.0
	5						8	78		
6	68.7396829368334	0.0	571.0	1.0	34.0	0.0	227817.778055783	3.43492073420835	134.0	1.0
	3						4	7		
7	65.0	0.0	224.0	1.0	50.0	0.0	149000.0	1.3	137.0	1.0
8	80.0	0.0	898.0	0.0	25.0	0.0	149000.0	1.1	144.0	1.0
9	66.0	1.0	72.0	0.0	40.0	1.0	242000.0	1.2	134.0	1.0
10	75.0	1.0	582.0	0.0	30.0	0.0	225000.0	1.83	134.0	1.0

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	50.624824690134889	0.0	563.0	0.0	48.0	0.0	153374.89481408094	0.6291584855396283	133.0	0.0
2	60.187392970024307	0.0	79.0	0.0	19.0	0.0	74230.087455711882	2.3109455272481772	134.0	1.0
3	56.698228861596377	0.0	2028.0	1.0	28.0	0.0	389253.3204275486	1.3733397862257079	136.0	1.0
4	69.428113249408601	0.0	123.0	0.0	44.0	1.0	283771.24529976345	1.2942811324940862	136.0	1.0
5	63.013211446383534	0.0	120.0	0.0	25.0	0.0	65986.788553616405	2.2080638003903523	134.0	1.0
6	54.290547120621284	0.0	302.0	0.0	43.0	0.0	153000.0	3.3888556284038347	134.0	0.0
7	55.0	0.0	572.0	1.0	35.0	0.0	231000.0	0.8	143.0	0.0
8	60.0	0.0	68.0	0.0	20.0	0.0	119000.0	2.9	127.0	1.0
9	70.0	0.0	97.0	0.0	60.0	1.0	220000.0	0.9	138.0	1.0
10	53.0	1.0	1808.0	0.0	60.0	1.0	249000.0	0.7	138.0	1.0

B.10 Audit Trace: Fold 10 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	62.971623423041535	1.0	391.0	1.0	39.0	0.0	196377.29873843322	1.2782974053824918	134.0	0.0
2	47.817123916091263	0.0	168.0	0.0	26.0	1.0	273271.404895114	1.6457190209771928	129.0	0.0
3	73.568199083844389	0.0	102.0	0.0	36.0	0.0	223578.88208348365	2.3905093417057475	134.0	1.0
4	91.952539223499377	0.0	582.0	0.0	38.0	0.0	263358.03	1.83	134.0	0.0
5	80.448687640480103	0.0	363.0	0.0	48.0	0.0	48057.712972399991	1.3282056792639976	135.0	0.0
6	81.7353396776918	0.0	334.0	0.0	56.0	0.0	306870.57608594885	1.1891177989256394	132.0	1.0
7	60.0	0.0	897.0	1.0	45.0	0.0	297000.0	1.0	133.0	1.0
8	75.0	0.0	582.0	0.0	20.0	1.0	265000.0	1.9	130.0	1.0
9	45.0	1.0	1876.0	1.0	35.0	0.0	226000.0	0.9	138.0	1.0
10	75.0	1.0	246.0	0.0	15.0	0.0	127000.0	1.2	137.0	1.0